Collaborative Wind Tunnel Flight Investigation

Multi-Variable Functional Scaling of Aerodynamic Stall

$$\alpha_{\text{stall}} = K \cdot (t/c)^a \cdot (1+m)^b \cdot Re^c$$

Platform: NCSSM *WindTunnel lab* (GPU-Accelerated 2D Lattice Boltzmann Simulator)

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Mission Briefing & Research Objective

In aerodynamics, **stall** represents the sudden loss of lift when airflow separates from the upper surface of a wing at a critical angle of attack α_{stall}. Using **Dimensional Analysis**, we can show that the critical stall angle depends on three fundamental dimensionless parameters: airfoil thickness ratio (t/c), camber ratio (m), and the Reynolds number (Re):

$$\alpha_{\text{stall}} = \Phi\left(\frac{t}{c},\, m,\, Re\right) \approx K \cdot \left(\frac{t}{c}\right)^a \cdot (1+m)^b \cdot Re^c. \tag{1}$$

As an aerospace research consortium, our class will systematically explore this multi-variable relationship using the GPU-accelerated **WindTunnel lab**. Three specialized student research teams—each dedicated to one governing dimensionless Π-group—will measure α_{stall} as a function of their assigned parameter, construct both the linear and log-log scaling plots to extract their power-law exponent (a, b, or c), and collaborate to calibrate and validate a unified empirical stall law.

1 Dimensional Analysis & The Aerodynamics of Stall

1.1 Physical Nature of Aerodynamic Stall

When an airfoil moves through air at an angle of attack α, fluid streams smoothly over both surfaces, generating lift. However, as α increases, air flowing over the upper surface must turn sharply around the nose and decelerate into an increasing pressure hill (known as an *adverse pressure gradient*). At a critical angle called the **stall angle** (α_{stall}), the sluggish air near the surface can no longer overcome this hill. The flow detaches (separates) from the upper surface, lift reaches its primary attached-flow peak and drops, and the wing stalls.

1.2 Formulating the Physical Problem with Dimensional Analysis

Before running expensive wind tunnel tests or computer simulations, aerospace engineers use **Dimensional Analysis** (the Buckingham Π Theorem) to determine which variables truly govern the physics, reducing complex systems of units into universal, dimensionless parameters.

To determine what governs the stall angle α_{stall}, we list every physical variable that could influence the onset of flow separation:

Physical Variable	**Symbol**	**SI Units**	**Dimensions** $[M, L, T]$
Stall Angle of Attack	α_{stall}	rad or deg	[1] (dimensionless)
Airfoil Chord Length	c	m	$[L]$
Maximum Thickness	t	m	$[L]$
Maximum Camber Height	h	m	$[L]$
Freestream Air Speed	U_∞	m/s	$[LT^{-1}]$
Air Density	ρ	kg/m^3	$[ML^{-3}]$
Air Dynamic Viscosity	μ	Pa · s	$[ML^{-1}T^{-1}]$

1.3 Applying the Buckingham Π Theorem

We have $n = 7$ physical variables expressed in $k = 3$ fundamental dimensions: mass $[M]$, length $[L]$, and time $[T]$. According to the Buckingham Π Theorem, the physical relationship can be expressed by exactly $n - k = 7 - 3 = 4$ independent dimensionless groups (Π-groups).

Selecting the fluid dynamic reference scale (ρ, U_∞, c) as repeating variables, we construct the dimensionless Π-groups:

1. **Target Dimensionless Output ($\Pi_\alpha = \alpha_{\textbf{stall}}$):** The critical stall angle of attack (dimensionless).
2. **Group 1 ($\Pi_1 = t/c$):** The **Airfoil Thickness Ratio** (maximum thickness relative to chord).
3. **Group 2 ($\Pi_2 = h/c = m$):** The **Airfoil Camber Ratio** (maximum camber relative to chord).
4. **Group 3 ($\Pi_3 = \frac{\rho U_\infty c}{\mu} = Re$):** The **Reynolds Number**, which scales fluid inertia (momentum) relative to viscous friction.

Therefore, dimensional analysis proves that the stall angle is strictly governed by three independent dimensionless groups:

$$\alpha_{\text{stall}} = \mathbf{\Phi}\left(\mathbf{\Pi_1}, \mathbf{\Pi_2}, \mathbf{\Pi_3}\right) = \mathbf{\Phi}\left(\frac{\mathbf{t}}{\mathbf{c}}, \mathbf{m}, \mathbf{Re}\right) \tag{2}$$

These three independent Π-groups directly define the division of labor across our **three specialized student research teams**.

1.4 Physical Intuition Behind the Dimensionless Groups

- **Airfoil Thickness (t/c):** Controls the roundness of the nose (leading-edge radius $r_{\text{LE}}/c = 1.1019\,(t/c)^2$). A sharp, thin airfoil ($t/c = 0.06$) forces air around an abrupt corner, causing early leading-edge separation. A thicker airfoil ($t/c = 0.18$) softens the leading-edge suction spike, delaying separation to higher angles, until excessive thickness ($t/c = 0.24$) triggers trailing-edge pressure-recovery stall.
- **Airfoil Camber (m):** Positive camber arches the mean line so the leading edge aligns more smoothly with the incoming airflow at positive incidence, delaying separation to slightly higher angles according to $\alpha_{\text{stall}} \propto (1 + m)^b$.
- **Reynolds Number (Re):** Measures fluid inertia relative to viscous friction. High-Re air has higher near-wall momentum and thinner laminar boundary layers ($\delta/c \propto Re^{-1/2}$), enabling the boundary layer to withstand steeper adverse pressure gradients and delaying stall according to $\alpha_{\text{stall}} \propto Re^c$.

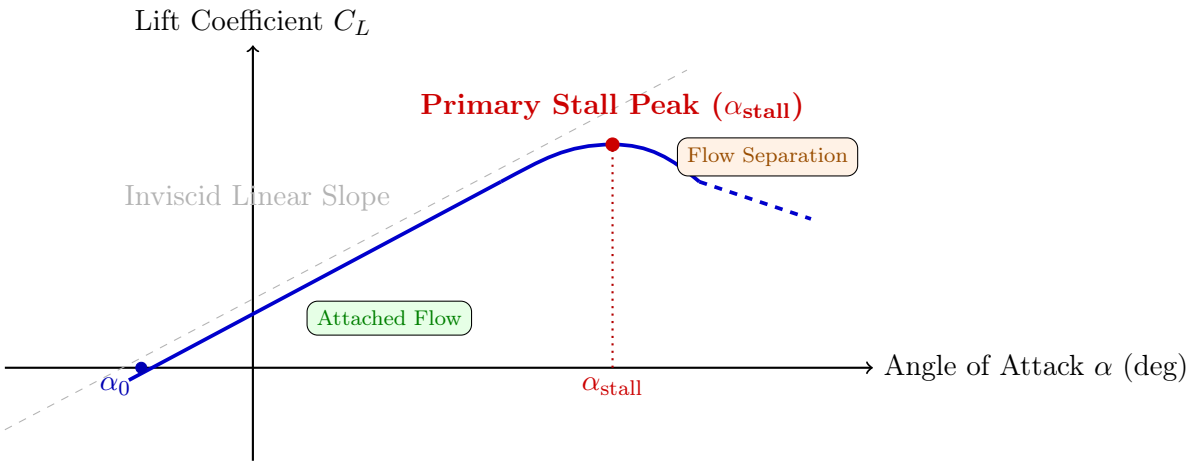


Figure 1: Identification of the critical stall angle of attack α_{stall} from an aerodynamic lift sweep.

2 Experimental Testing Platform: NCSSM *WindTunnel lab*

Tests are conducted using NCSSM's **WindTunnel lab**, a real-time GPU-accelerated numerical wind tunnel powered by the Lattice Boltzmann Method (LBM D2Q9) and Smagorinsky subgrid-scale turbulence modeling.

2.1 Standardized Data Acquisition (DAQ) Protocol

To ensure reproducibility across all student research teams, each angle sweep is executed using the **5-Second Cycle-Averaged DAQ Protocol**:

1. **Geometry & Flow Configuration:** Enter the 4-digit NACA profile (e.g., `2412`), set the target Reynolds number (Re), and verify that *Smagorinsky LES* is enabled.
2. **Angle Dialing & Flow Settling:** Sweep α using coarse steps (2.0°) in the linear attached region and fine steps (0.5°) across the stall zone. Wait 3–5 seconds at each angle until the status badge displays **Flow Settled · Ready**.
3. **5s Cycle-Averaged Acquisition:** Click `[Acquire (5s)]` (or press `L`) to integrate forces over complete vortex-shedding periods and identify the primary attached-flow stall angle α_{stall} for each profile.

3 Consortium Research Team Assignments & Deliverables

The class is partitioned into **three specialized research teams**, corresponding directly to the three dimensionless Π-groups derived from Buckingham Π analysis:

- **Team 1 ($\Pi_1 = t/c$):** Airfoil Thickness Ratio Team
- **Team 2 ($\Pi_2 = m$):** Airfoil Camber Ratio Team
- **Team 3 ($\Pi_3 = Re$):** Reynolds Number Team

For each team, the required graphical deliverables are strictly focused on **two plots**:

1. **Main Plot:** Stall angle of attack α_{stall} as a function of the team's governing parameter (linear scale) with the fitted power-law curve overlaid.
2. **Log-Log Plot:** The log-log transformation of that relationship used to extract the power-law exponent via linear regression.

3.1 Team 1: Airfoil Thickness Ratio ($\Pi_1 = t/c$) Team

Primary Objective: Measure α_{stall} as a function of thickness ratio t/c, extract the power-law exponent a in $\alpha_{\text{stall}} \propto (t/c)^a$, and measure the coupled thin-camber validation benchmark `NACA 4406`.

- **Fixed Controls:** Baseline Reynolds number $Re = 1000$, Camber $m = 0.00$ (Symmetric Series).
- **Primary Airfoil Sweep Matrix:** `NACA 0006` ($t/c = 0.06$), `NACA 0009` ($t/c = 0.09$), `NACA 0012` ($t/c = 0.12$), `NACA 0018` ($t/c = 0.18$), and `NACA 0024` ($t/c = 0.24$).
- **Team 1 Deliverables (Two-Panel Figure & Table):**
  1. **Main Plot (α_{stall} vs. t/c):** Plot measured stall angle α_{stall} (°) as a function of thickness ratio t/c, overlaid with the fitted power-law curve $\alpha_{\text{stall}} = A\,(t/c)^a$.
  2. **Log-Log Plot ($\ln(\alpha_{\text{stall}})$ vs. $\ln(t/c)$):** Plot the log-transformed coordinates for the attached-stall regime ($t/c \in [0.06, 0.18]$) and extract the thickness exponent a from the linear regression slope:
  $$\ln(\alpha_{\text{stall}}) = a\,\ln\left(\frac{t}{c}\right) + \ln A \quad \Longrightarrow \quad \alpha_{\text{stall}} \propto \left(\frac{t}{c}\right)^a, \tag{3}$$
  noting the high-thickness saturation departure at $t/c = 0.24$ (`NACA 0024`).
- **Coupled Validation Benchmark:** Measure α_{stall} on `NACA 4406` ($m = 0.04, t/c = 0.06, Re = 1000$).

3.2 Team 2: Airfoil Camber Ratio ($\Pi_2 = m$) Team

Primary Objective: Measure α_{stall} as a function of maximum camber ratio m, extract the power-law exponent b in $\alpha_{\text{stall}} \propto (1 + m)^b$, and measure the coupled thick-camber validation benchmark `NACA 4418`.

- **Fixed Controls:** Baseline Reynolds number $Re = 1000$, Thickness $t/c = 0.12$, Maximum Camber Position $p = 0.4$ (`NACA X412` Family).
- **Primary Airfoil Sweep Matrix:** `NACA 0012` ($m = 0.00$), `NACA 1412` ($m = 0.01$), `NACA 2412` ($m = 0.02$), `NACA 4412` ($m = 0.04$), and `NACA 6412` ($m = 0.06$).
- **Team 2 Deliverables (Two-Panel Figure & Table):**
  1. **Main Plot (α_{stall} vs. m):** Plot measured stall angle α_{stall} (°) as a function of maximum camber ratio m, overlaid with the fitted power-law curve $\alpha_{\text{stall}} = B\,(1+m)^b$.
  2. **Log-Log Plot ($\ln(\alpha_{\text{stall}}/\alpha_{\text{stall},0})$ vs. $\ln(1+m)$):** Plot the normalized log-transformed stall angle relative to the symmetric baseline $\alpha_{\text{stall},0}$ (`NACA 0012`) against $\ln(1 + m)$ across $m \in [0.00, 0.06]$, and extract the camber exponent b from the linear regression slope:
  $$\ln\left(\frac{\alpha_{\text{stall}}}{\alpha_{\text{stall},0}}\right) = b\,\ln(1 + m) \quad \Longrightarrow \quad \alpha_{\text{stall}} \propto (1 + m)^b. \tag{4}$$
- **Coupled Validation Benchmark:** Measure α_{stall} on `NACA 4418` ($m = 0.04, t/c = 0.18, Re = 1000$).

3.3 Team 3: Reynolds Number ($\Pi_3 = Re$) Team

Primary Objective: Measure α_{stall} as a function of Reynolds number Re on `NACA 2412` and extract the power-law exponent c in $\alpha_{\text{stall}} \propto Re^c$.

- **Fixed Controls:** Standard `NACA 2412` profile ($m = 0.02$, $p = 0.4$, $t/c = 0.12$).
- **Primary Reynolds Number Sweep Matrix:** $Re \in \{300, 600, 1200, 2500, 5000\}$.
- **Team 3 Deliverables (Two-Panel Figure & Table):**
  1. **Main Plot (α_{stall} vs. Re):** Plot measured stall angle α_{stall} (°) as a function of Reynolds number Re, overlaid with the fitted power-law curve $\alpha_{\text{stall}} = C\,Re^c$.
  2. **Log-Log Plot ($\ln(\alpha_{\text{stall}})$ vs. $\ln(Re)$):** Plot $\ln(\alpha_{\text{stall}})$ against $\ln(Re)$ across all five Reynolds numbers and extract the Reynolds exponent c from the linear regression slope:

$$\ln(\alpha_{\text{stall}}) = c\,\ln(Re) + \ln C \quad \Longrightarrow \quad \alpha_{\text{stall}} \propto Re^c. \tag{5}$$

4 Classroom Consortium: Multi-Variable Data Pooling

Following data acquisition, each team records its measured stall angles α_{stall} and log-transformed coordinates in the **Consortium Master Stall Table** (Table 1) to synthesize and validate the unified stall law.

Table 1: Consortium Master Stall Angle Telemetry Table organized by Π-group research teams.

Team	Airfoil / Case	m	t/c	Re	Measured α_{stall} (°)	Log-Transformed $\ln(\alpha_{\text{stall}})$
Team 1: Thickness ($\Pi_1 = t/c$) Matrix ($m = 0.00$, $Re = 1000$)						
1	NACA 0006	0.00	0.06	1000	_____	_____
1	NACA 0009	0.00	0.09	1000	_____	_____
1	NACA 0012 (Symmetric Ref.)	0.00	0.12	1000	_____	_____
1	NACA 0018	0.00	0.18	1000	_____	_____
1	NACA 0024	0.00	0.24	1000	_____	_____
Team 2: Camber ($\Pi_2 = m$) Matrix ($t/c = 0.12$, $Re = 1000$) — report $\ln(\alpha_{\text{stall}}/\alpha_{\text{stall},0})$						
2	NACA 0012 (Baseline $\alpha_{\text{stall},0}$)	0.00	0.12	1000	_____	0.0000
2	NACA 1412	0.01	0.12	1000	_____	_____
2	NACA 2412	0.02	0.12	1000	_____	_____
2	NACA 4412	0.04	0.12	1000	_____	_____
2	NACA 6412	0.06	0.12	1000	_____	_____
Team 3: Reynolds Number ($\Pi_3 = Re$) Matrix (NACA 2412, $m = 0.02, t/c = 0.12$)						
3	NACA 2412 ($Re = 300$)	0.02	0.12	300	_____	_____
3	NACA 2412 ($Re = 600$)	0.02	0.12	600	_____	_____
3	NACA 2412 ($Re = 1200$)	0.02	0.12	1200	_____	_____
3	NACA 2412 ($Re = 2500$)	0.02	0.12	2500	_____	_____
3	NACA 2412 ($Re = 5000$)	0.02	0.12	5000	_____	_____
Out-of-Sample Coupled Multi-Parameter Validation Benchmarks ($Re = 1000$)						
1	NACA 4406 (Thin + High Camber)	0.04	0.06	1000	_____	_____
2	NACA 4418 (Thick + High Camber)	0.04	0.18	1000	_____	_____

5 Analysis & Synthesis Requirements

Each research team will present its two-panel deliverable figure and data table, followed by the consortium synthesis of the unified stall scaling law:

1. **Team 1 Deliverable ($\Pi_1 = t/c$):** Present the data table of $(t/c, \alpha_{\text{stall}}, \ln(t/c), \ln(\alpha_{\text{stall}}))$, the main plot of α_{stall} vs. t/c, and the log-log plot $\ln(\alpha_{\text{stall}})$ vs. $\ln(t/c)$. Report the extracted exponent a and explain why `NACA 0024` ($t/c = 0.24$) saturates below the thin-to-moderate thickness power-law trend.

2. **Team 2 Deliverable ($\Pi_2 = m$):** Present the data table of $(m, 1+m, \alpha_{\text{stall}}, \ln(1+m), \ln(\alpha_{\text{stall}}/\alpha_{\text{stall},0}))$, the main plot of α_{stall} vs. m, and the log-log plot $\ln(\alpha_{\text{stall}}/\alpha_{\text{stall},0})$ vs. $\ln(1+m)$. Report the extracted camber exponent b.

3. **Team 3 Deliverable ($\Pi_3 = Re$):** Present the data table of $(Re, \alpha_{\text{stall}}, \ln(Re), \ln(\alpha_{\text{stall}}))$, the main plot of α_{stall} vs. Re, and the log-log plot $\ln(\alpha_{\text{stall}})$ vs. $\ln(Re)$. Report the extracted Reynolds number exponent c and relate it to laminar boundary-layer thinning ($\delta/c \propto Re^{-1/2}$).

4. **Consortium Synthesis & Coupled Validation:** Combine the three empirically extracted exponents (a, b, c) into the unified general stall law:

$$\alpha_{\text{stall}}\left(\mathbf{m},\, \frac{\mathbf{t}}{\mathbf{c}},\, \mathbf{Re}\right) = \mathbf{K} \cdot \left(\frac{\mathbf{t}}{\mathbf{c}}\right)^{\mathbf{a}} \cdot (\mathbf{1}+\mathbf{m})^{\mathbf{b}} \cdot \mathbf{Re}^{\mathbf{c}}, \tag{6}$$

   determine the calibration constant K, and validate the formula's predictions (α_{pred}) against the experimentally measured stall angles (α_{meas}) on the two out-of-sample coupled benchmarks `NACA 4406` ($m = 0.04, t/c = 0.06, Re = 1000$) and `NACA 4418` ($m = 0.04, t/c = 0.18, Re = 1000$):

$$\text{Percent Error} = \left|\frac{\alpha_{\text{pred}} - \alpha_{\text{meas}}}{\alpha_{\text{meas}}}\right| \times 100\%. \tag{7}$$

6 Grading Rubric (100 Points Total)

Criterion	Performance Expectations	Points
Experimental Protocol & DAQ Rigor	Correct execution of the 5s cycle-averaged DAQ protocol; flow-settled state verified; 0.5° angular resolution near stall; complete stall angle tables.	25
Team Deliverable Plots (Main & Log-Log)	Clear two-panel figures for each team: (a) α_{stall} as a function of the governing parameter with power-law fit curve, and (b) log-log version with linear regression slope and R^2.	30
Consortium Synthesis & Validation	Accurate assembly of exponents (a, b, c) and calibration constant K into $\alpha_{\text{stall}} = K(t/c)^a(1+m)^b Re^c$; validation and percent error calculation on `NACA 4406` and `NACA 4418`.	30
Technical Discussion & Formatting	Concise physical discussion of the scaling exponents and high-thickness saturation; clean report formatting and tables.	15
Total		**100**

Table 2: Laboratory Evaluation and Assessment Rubric.